

```
import numpy as np
import pandas as pd
from sklearn.ensemble import RandomForestRegressor

def analyze_market_data(data):
    # Clean and structured approach
    cleaned_data = data.dropna()
    features = cleaned_data[['Open', 'High', 'Low', 'Volume']]
    target = cleaned_data['Close']

    model = RandomForestRegressor(n_estimators=100, random_state=42)
    model.fit(features, target)

    predictions = model.predict(features)
    return predictions, model.feature_importances_
```

```
[1.002  0.985  1.012  5000]
[1.015  1.000  1.030  4800]
[1.008  0.995  1.020  5200]
[1.020  1.005  1.035  5100]
[1.012  0.998  1.025  4900]
[1.012  0.998  1.032  5300]
```

When AI Meets Market Reality

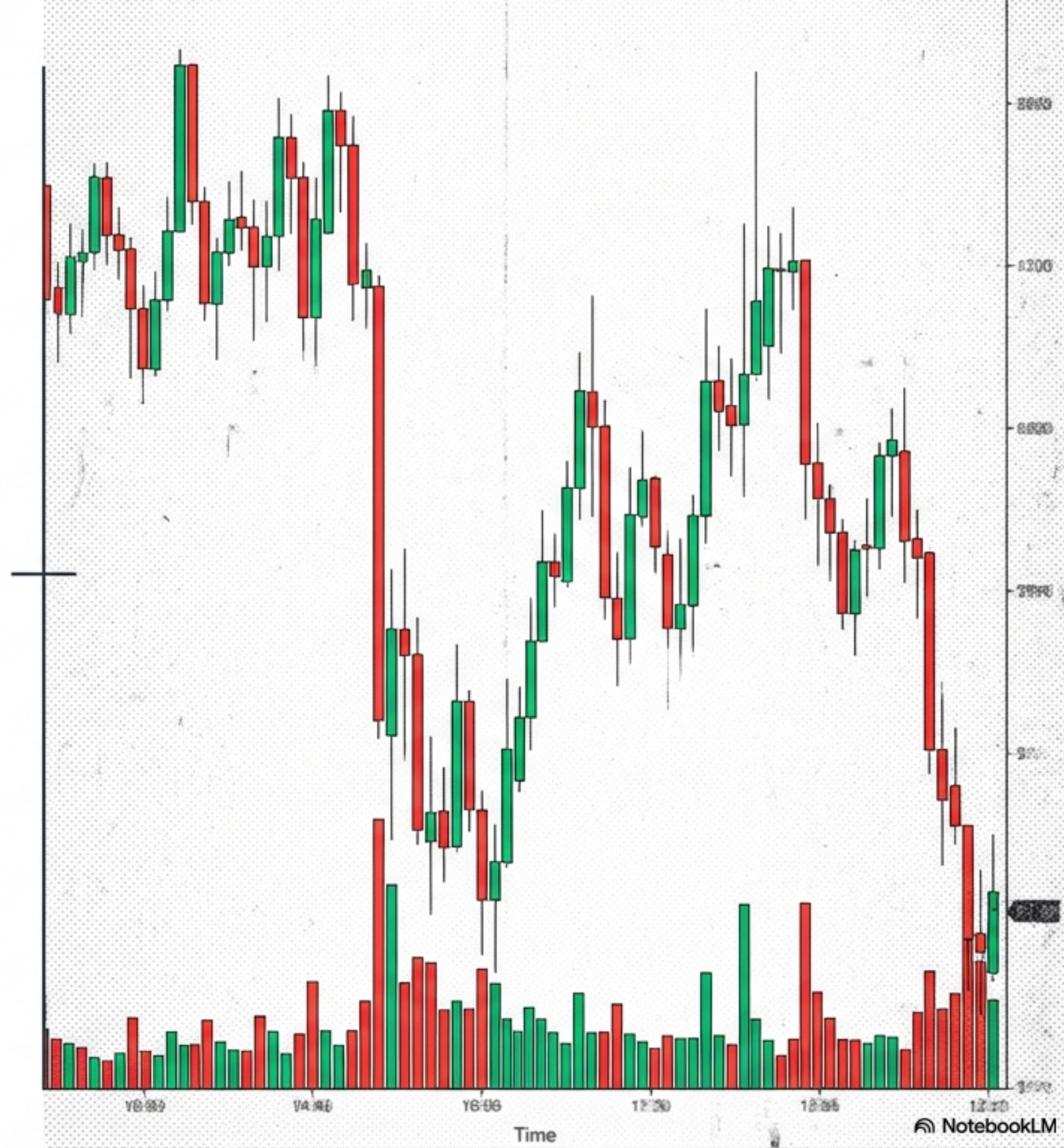
Midterm Presentation: VTS Tournament - Stacking Sats Challenge

Key Insight:

Midterm Finding: Baseline Beats CNN (So Far)

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Practicum Midterm Presentation | February 2026



Executive Summary: The Scientific Pivot



The Trap

Built a 296K-parameter Deep CNN to decode market noise based on academic literature.



The Reality

Model underperformed a random coin flip.

41.43% Percentile (CNN) vs **41.94%** (Baseline).



The Success

Validated causality, built bulletproof ablation harness, and identified root causes.

Infrastructure & Rigor: 



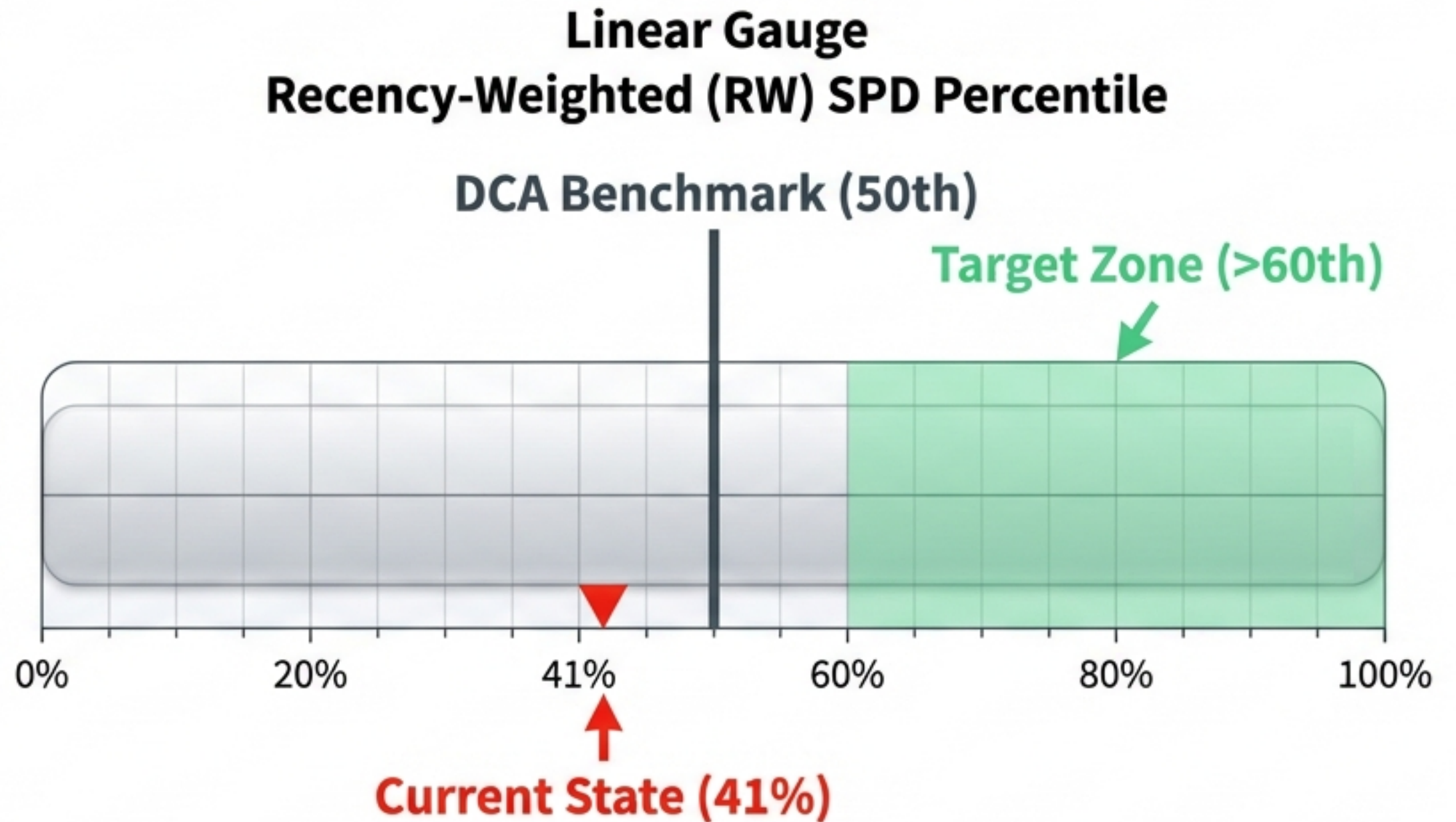
The Pivot

Shifted to Simplified Baseline. Implementing Regime Detection & Weekly Granularity.

Pivot Strategy:  In Progress

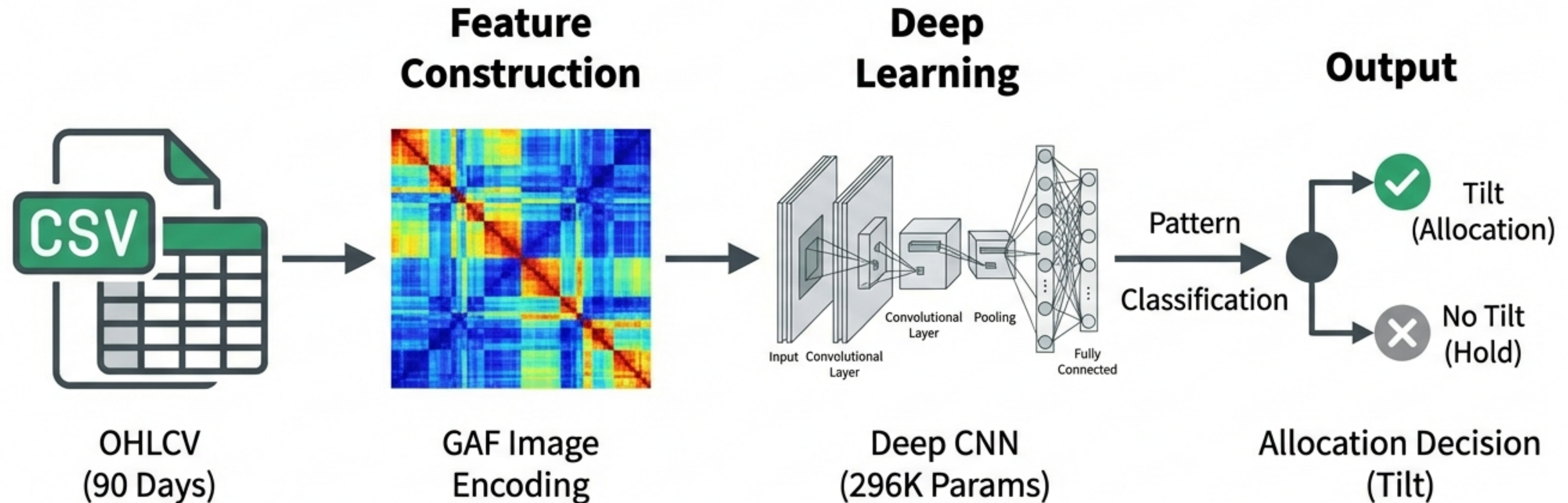
The Challenge: Beat DCA Under Strict Causality

- **Objective:** Predict daily BTC allocation weights (2016-2025).
- **Constraints:** Strict Causality (No lookahead),
 $\sum w_i = 1.0$,
 $w_i \geq 1e-5$.



The Hypothesis: Visual Patterns Predict Price Action

Inspiration: Classic Technical Analysis (Magee & Edwards) + Previous JPM Research (0.90+ F1 Scores).

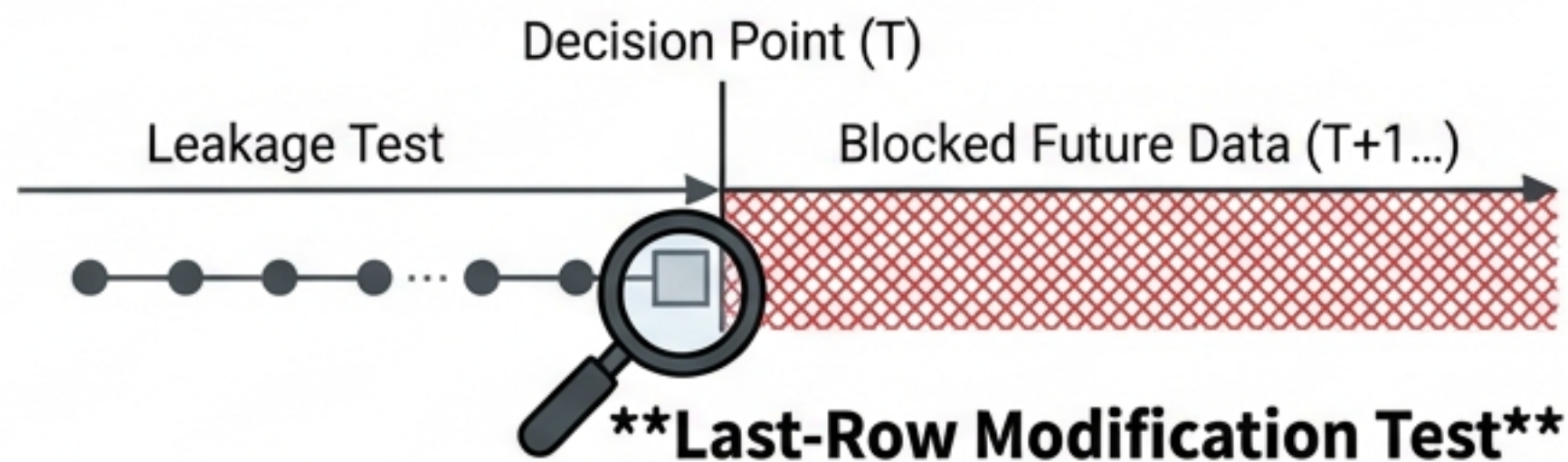


Methodology: Ensuring Scientific Rigor

Data Pipeline

- **Dataset:** BTC Daily OHLCV (2016-2025) via Polygon.
- **Training:** 2014-2015 (Pre-tournament).
- **Evaluation:** 3,076 rolling walk-forward windows.

The "Shield" (Causality Verification)



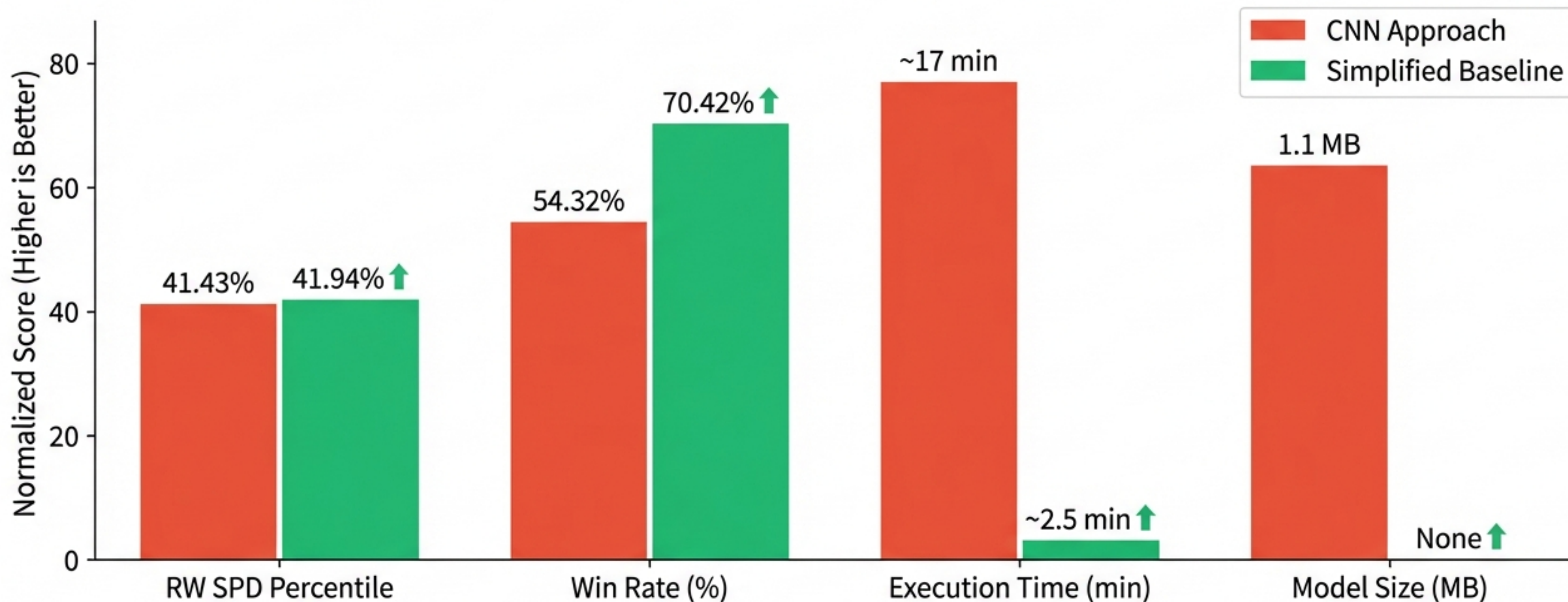
Difference: **0.00e+00**

Status: **Zero Leakage Confirmed**

Strict temporal embargo implemented to guarantee genuine failure.

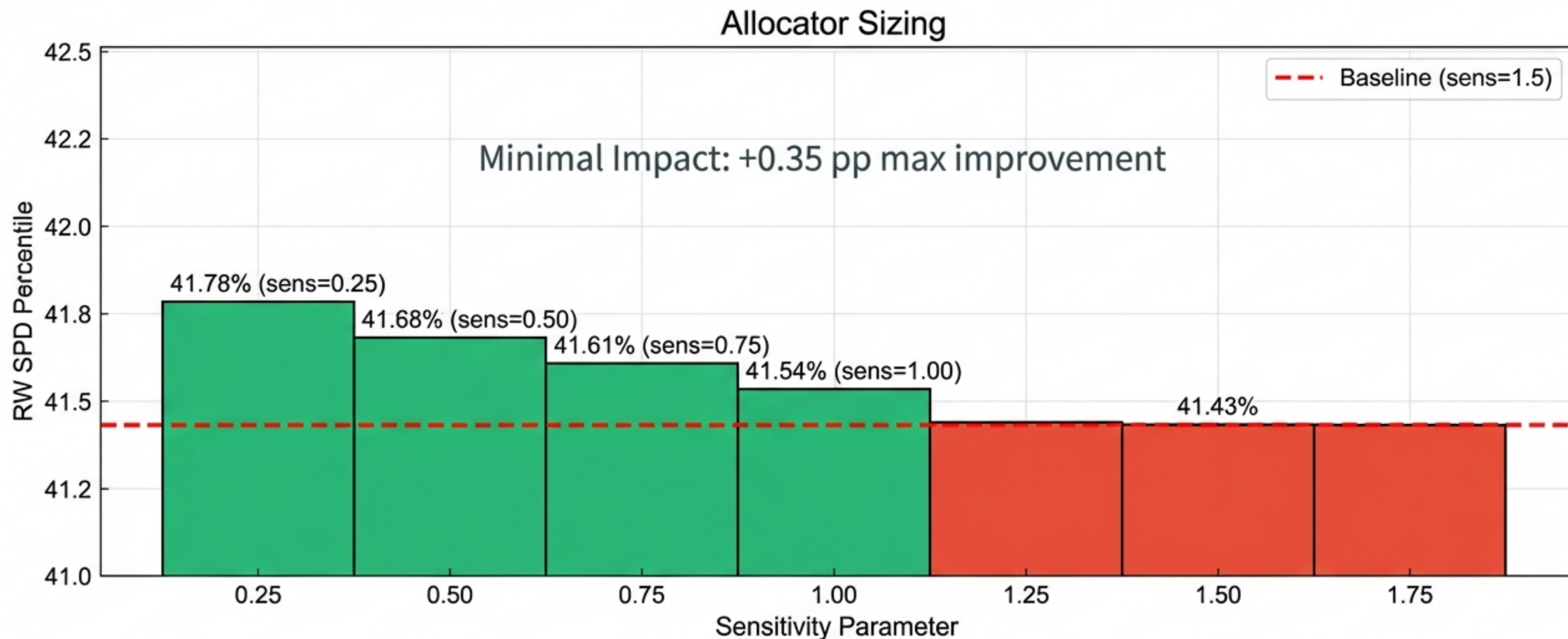
The Reality Check: Complexity Failed to Perform

Performance Comparison: CNN vs Simplified Baseline (Simplified Wins on ALL Dimensions)



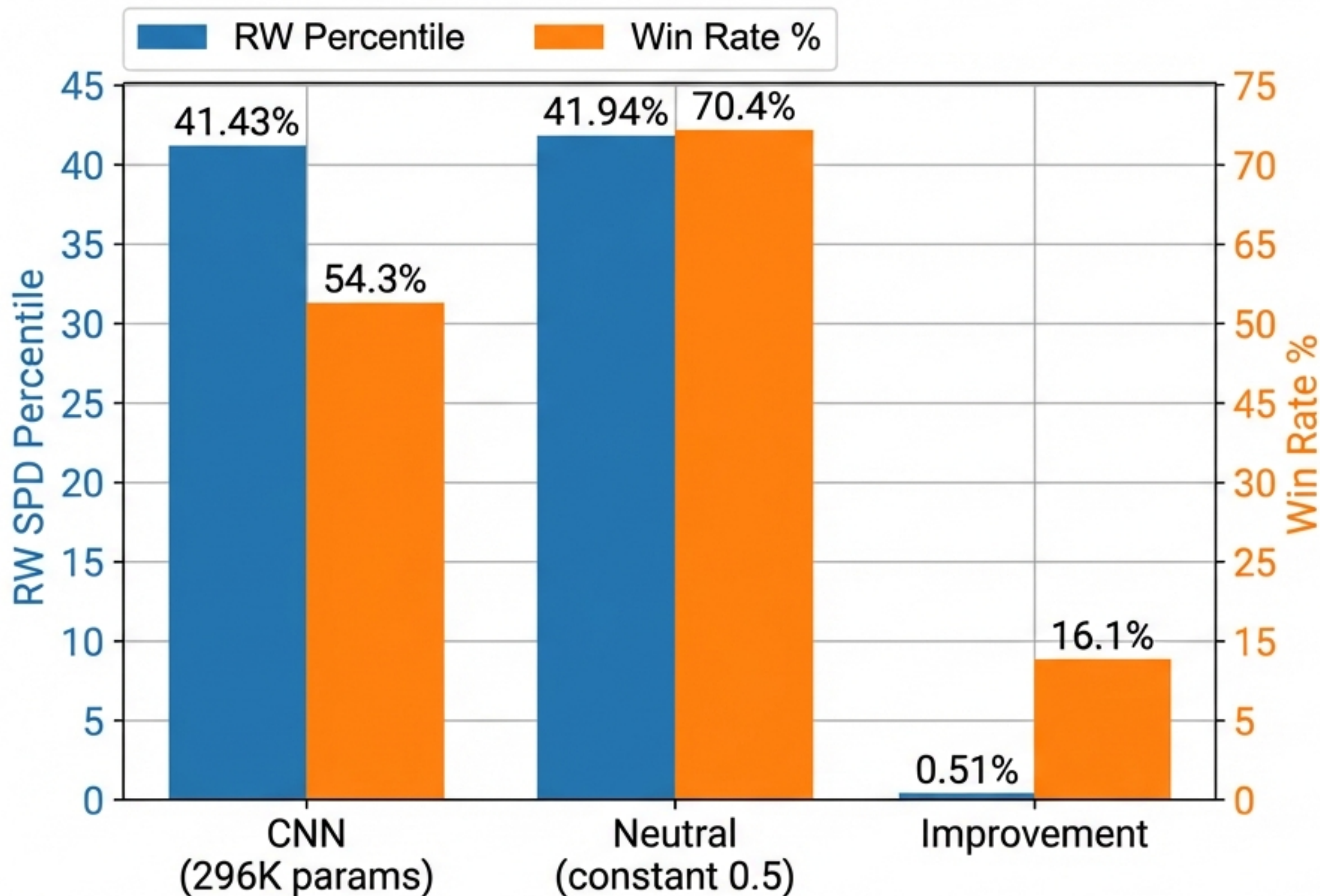
The complex approach loses on all dimensions: Score, Consistency, Speed, and Efficiency.

Ablation Study #1: Was it the Bet Size?



The problem isn't the *magnitude* of the bets (allocator aggressiveness); it is the *direction*.

Ablation Study #2: The Signal Quality Test (The “Aha!” Moment)

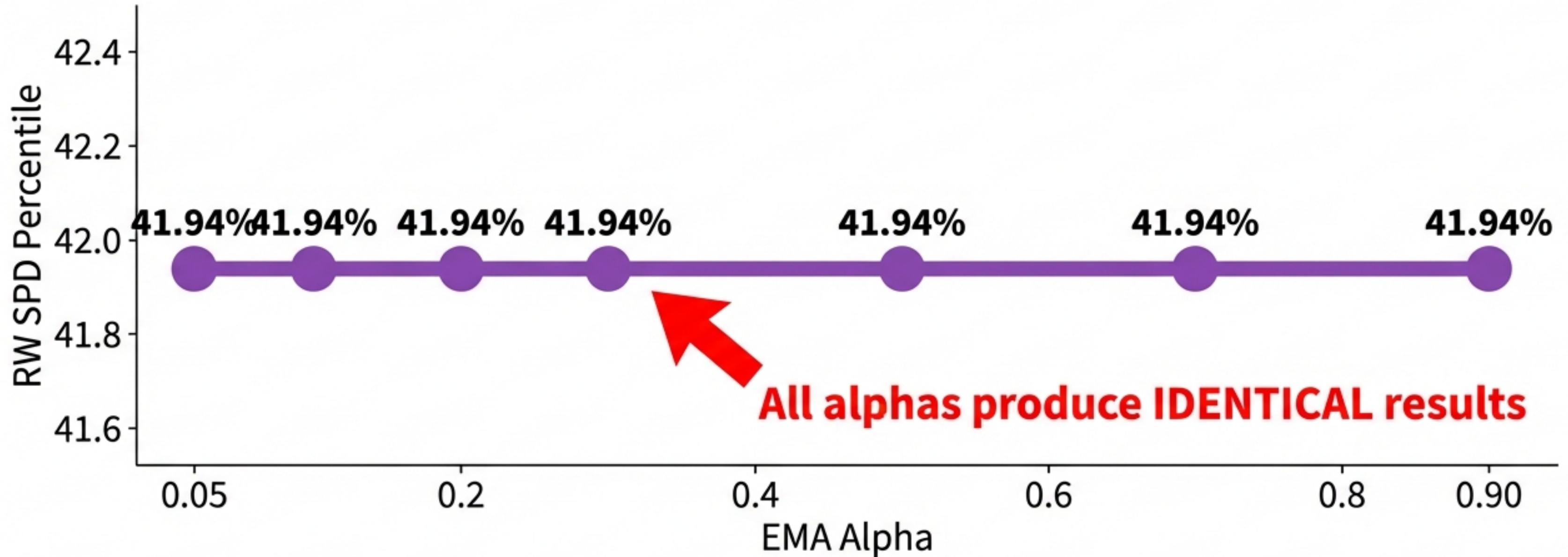


****CNN $\equiv$ Random Guessing****

The neutral signal improved Win Rate by 16.1%.

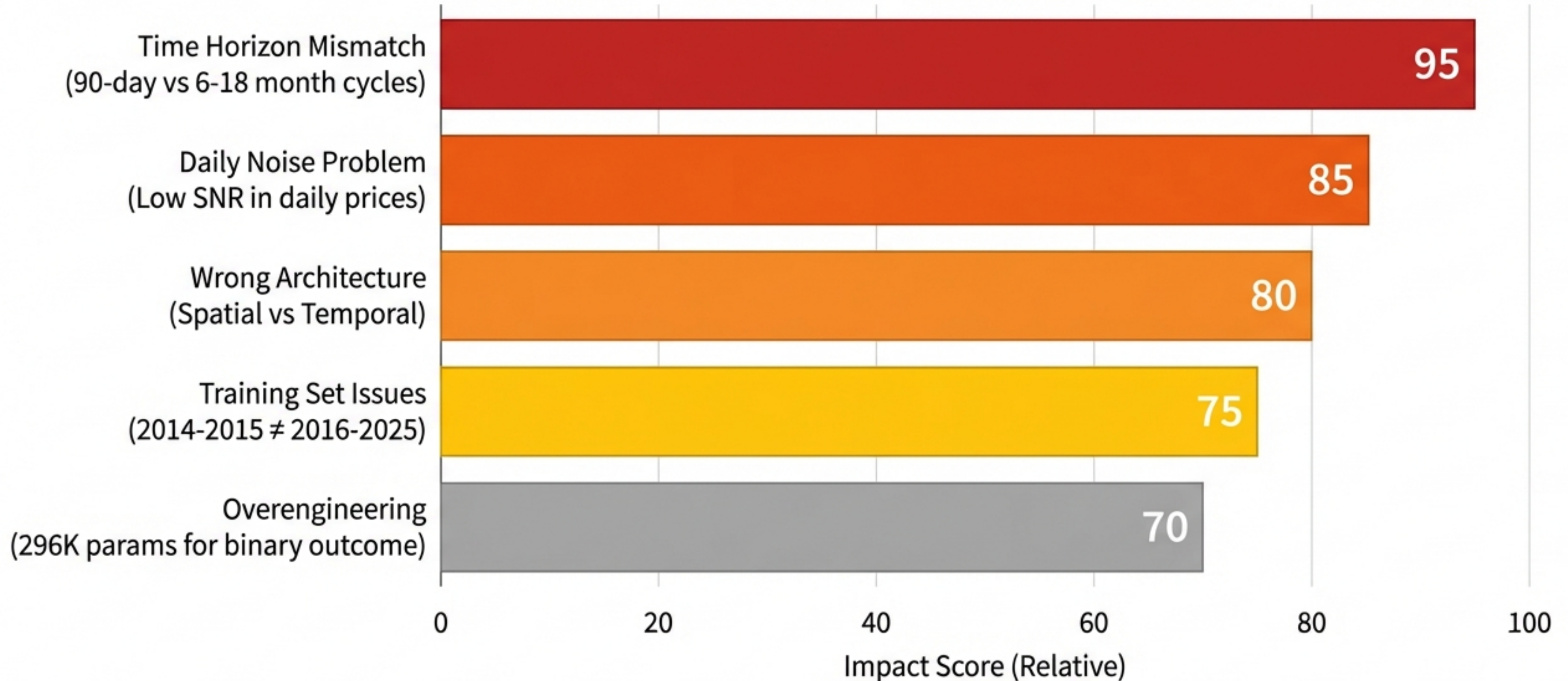
Ablation Study #3: The Limits of Smoothing

EMA Smoothing Parameter Sweep



Mathematical Proof: When the signal is constant/noise, the tilt is zero. You cannot smooth a flat line into a profit curve.

Root Cause Analysis: Why GAF + CNN Failed



The Pivot: Embracing Occam's Razor

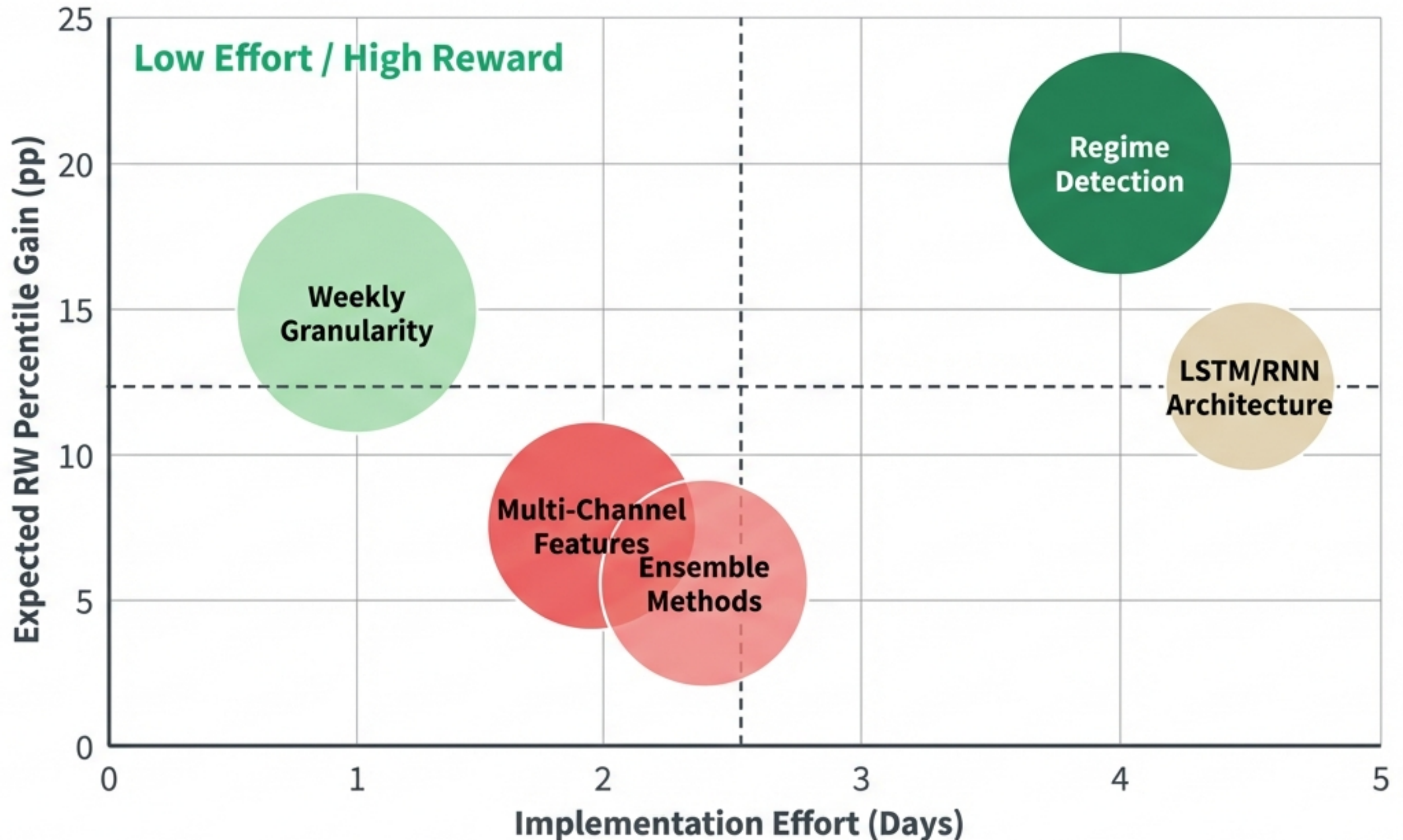
The Simplified Baseline

- **Core Logic:** Constant `prob_up = 0.5` (Neutral Probability).
- **Architecture:** Tilt → Bounded Multiplier → EMA (No Neural Net).
- Performance: **70.42% Win Rate** (vs 54% for CNN).
- Model Size: **0 MB**.

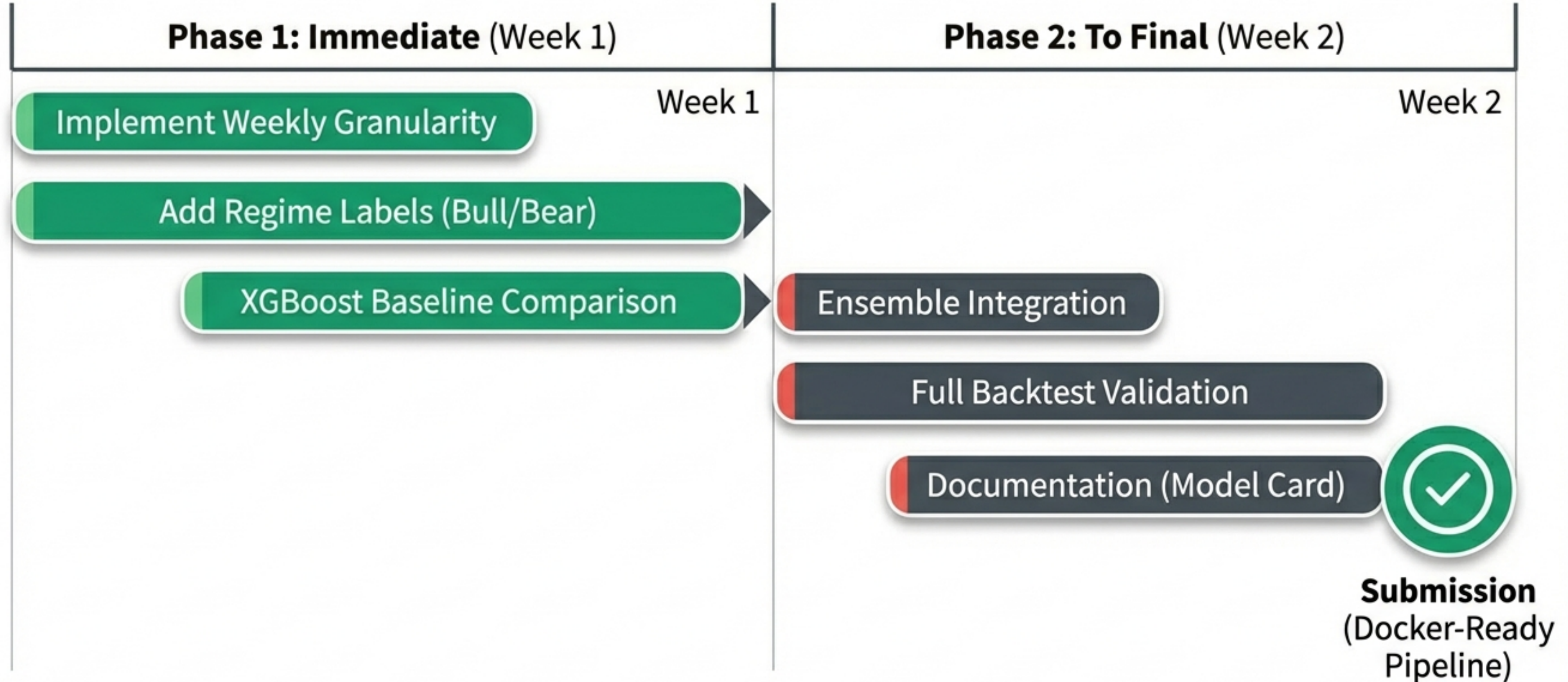
"The best model is the one that works, not the one that sounds impressive."

Path to Final: ROI Analysis of Future Improvements


Target: Move RW
Percentile from
Percentile from
41.94% → 57-73%.



Timeline & Deliverables



Lessons Learned: Scientific Integrity > Raw Performance

1. Ablation is Critical.
Without it, we'd never know the CNN was worthless.

2. Test "Random" Early.
A coin flip should be Baseline #0, not Ablation #2.

3. Complexity ≠ Performance.
296K parameters lost to a neutral constant.

4. Simplicity Wins.
In financial time series, signal quality matters more than model architecture.

Source Sans Pro Regular, Source-Pro Italic, Slate Charcoal

"We falsified a plausible deep-learning approach under strict causality. That is a success."

Discussion & Feedback Requests

1.

Metric Interpretation

Is our understanding of the Recency-Weighted SPD percentile correct?

2.

Split Strategy

Is the 2014-2015 (Train) / 2016-2025 (Test) split split appropriate, or should we use expanding windows?

3.

Pivot Scope

Is the plan for “Weekly + Regime Detection” appropriately scoped for the remaining timeline?

4.

Baseline Choice

Should we maintain the simplified constant as primary, or invest more in an XGBoost ensemble?

Repository: Visual_Trading_System

Notebook: btc_accumulation_model_simplified.ipynb